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Roll No

CE-505 (GS)

B.E. V Semester Examination, June 2020

Grading System (GS)

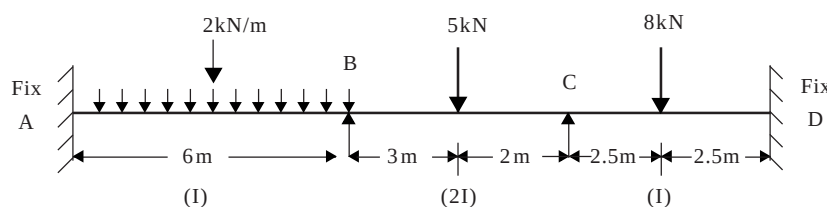
Theory of Structure - I

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. What are the different types of the strain energies stored in the structure?
2. Find out the expressions for horizontal reactions, cable tensions at the ends and shape of the cable for a uniformly loaded cable.
3. State principle of virtual work. Explain its application to flexural members. Find vertical deflection and rotation at the end of cantilever beam of span L subjected to a concentrated load P at the free end. EI is constant.
4. Analyse the beam by slope deflection method or moment distribution method. Plot the B.M. Diagram and Deflected shape.



5. What do you understand by strain energy? Also explain Maxwell's reciprocal theorem.

OR

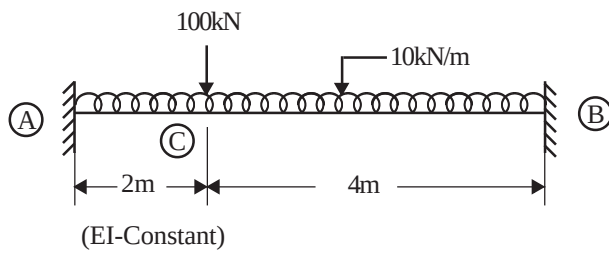
What do you understand by influence line diagram? State the significance of ILD in the analysis of structures.

6. Describe static and kinematic indeterminacy. Also define stiffness at any frame.
7. Define:
 - i) Distribution factor
 - ii) Rolling loads
 - iii) Effect at sinking at supports
 - iv) Non sway at frames

OR

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Analyze the Beam shown in Fig by Column Analogy Method.



8. Write short note (any two):

- a) Eddy's theorem
- b) Influence lines, Muller-Breslau principle
- c) Deformation of cable
